

Efter hver blok udfylder kursusansvarlig dette spørgeskema der giver anledning til generel refleksion over kurset. Evalueringen er et vigtigt redskab for Undervisningsudvalgets arbejde med kvalitetssikring af undervisningen og de forskellige kurser.

Institut der udbyder kurset i blok 3 2025/26:

- ☐ Biologisk institut (BIO)
- ☐ Datalogisk institut (DIKU)
- ☐ Institut for Fødevare- og Ressource Økonomi (IFRO)
- ☐ Institut for Fødevarevidenskab (FOOD)
- ☐ Institut for Geovidenskab og Naturforvaltning (IGN)
- ☐ Institut for Idræt og Ernæring (NEXS)
- ☐ Institut for Matematiske Fag (MATH)
- ☐ Institut for Naturfagenes Didaktik (IND)
- ☐ Institut for Plante- og Miljøvidenskab (PLEN)
- ☐ Kemisk Institut (CHEM)
- ☒ Niels Bohr Institutet (NBI)
- ☐ Statens Naturhistoriske Museum (SNM)

Kursuskode og -navn – Findes på kurser.ku.dk

Eksempel på kursuskode: NIFB13002U

Eksempel på kursusnavn: Videnskabsteori for husdyrvidenskab

Det skal være præcis det, der fremgår i kursusbeskrivelsen.

Kursuskode NDAB23002U

Kursusnavn Introduktion til diskret matematik og algoritmer (IDMA)

Notatet er skrevet af

Jakob Nordström

Reflekterende og kritisk gennemgang af de studendes skriftlige evaluering, både spørgsmål 1-9 og fritekstkommentarer. Du bedes opdele din refleksion over begge dele i positiv feedback og konstruktiv kritik.

Positiv feedback	As a general comment, the IDMA course is considered to be demanding, covering a wide array of topics, and to require substantial work. However, students also tend to feel that the course is stimulating and useful, and that it is well-organized. The evaluation of the course last year was positive, but this year is even better. The scoring of various aspects of the course is clearly higher across the board compared to last year. Above 50% of students "strongly agree" the course has been useful and above 80% if "agree" is also included. Students write that the lectures were good and engaging. Working on
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problem sets was helpful, and prepared students well for the exam (since the consist of old exam problems). The exercise sessions with the teaching assistants were very, very good. Overall, the different components of the course fit well together. This is clear both from questions asking students to score different aspects and from the free-text answers. In particular, the free-text comments point out how important the TAs and exercises classes are, and the overall assessment of the TAs is even more positive than last year. Jakob Nordström is assessed as an excellent teacher overall, but also as quite demanding. Previous years, there were some concerns about the atmosphere sometimes being intimidating, but such comments have more or less completely disappeared. Regarding the exam, the students were very happy with the exam problems last year but less so this year. In the end, the exam probably turned out a bit harder than last year --- this is the assessment both of myself and (independently) of several of the TAs. It is all the more encouraging, then, that the results on the exam were overall clearly better than last year, with an all-time-low failure rate. The course evaluation has a fairly substantial part with additional questions. The main takeaways from this part concern the compulsory problem sets that students have to hand in throughout the course. Even though it is possible to solve the problem sets more or less perfectly with generative AI, this is not what the students are doing. Early on during the course, it was emphasized in the lectures and on Absalon that students should work on their own to learn, and these admonitions seem to have had the desired effect. Most students worked on the problems mostly on their own (which we can tell fairly confidently, since the results would have been much, much better if they would just have fed the problems to their favourite chatbot). Interestingly, it seems that students agree not to use chatbots as long as the workload is perceived to be reasonable and fair, but not otherwise. On one of the problem sets we had an extra hard problem that was not clearly marked as optional. For that problem, we could see clearly that many students had just copied the chatbot answer. So the implicit deal between teacher and students seems to be that as long as the teacher assigns reasonable problems, the students will be happy to make an honest attempt, but if the problems are impossibly hard, then students retaliate by unleashing the full

	power of generative AI. Another point reinforced in the answers to the additional questions is that the exercise classes are very valuable, to the extent that several students say that the exercise classes were where they really learned the course material.
Konstruktiv kritik	As already mentioned, Jakob Nordström is assessed as an excellent teacher overall, but also as quite demanding. The handwriting on the blackboard can be hard to read. Students are also concerned about time management during the lectures, where it is not always the case that everything in the lecture notes gets covered (though this is sometimes by design), and that lectures sometimes go over time. Last year, it was emphasized throughout the lectures how important it is to study hard on this course, since the material is challenging. This was then perceived as intimidating by some students, and there were comments that such perceptions could also make it hard to ask questions. Very happily, these concerns have mostly disappeared. The same message was conveyed to the students this year, but apparently much more successfully. Some students express that the amount of material covered on the course is too large. This might be true to some extent, but this is also what the study programs have requested. Overall, the student assessment of the 2025/26 edition of IDMA was clearly more positive than for 2024/25. There were also clear improvements in exam results for the second year in a row. To a large extent, this might be thanks to the teaching assistants, who we know play a crucial role for the students, and where we seem to have been lucky to get an excellent team of students this year.

Din mundtlige evaluering med studerende:

- ☐ Hvorledes bidrog den mundtlige evaluering til uddybning og forklaring af nogle af de væsentlige forhold i den skriftlige evaluering (uddyb)

- ☒ Jeg har ikke lavet en mundtlig evaluering

Her bedes den kursusansvarlige (du) kort forholde sig til væsentlige kritikpunkter fra evalueringen. Det kan fx være ved at beskrive, hvordan eventuelle problemer kan imødekommes fremover – eller hvorfor visse forhold ikke kan ændres. Det er også relevant at nævne positive elementer i kurset og evt. planer for at fastholde disse. Prioriter gerne det vigtigste.

IDMA is perceived as a demanding course, but I believe this is unavoidable. For most students, this is the first time in their life that they encounter rigorous mathematical reasoning. This is something that is very, very challenging for many students, and that requires lots of work (and lots of time) to get used to. But precisely because this is challenging, this is also something that the students need to learn. The ability to combine creative problem-solving skills with logic reasoning is probably the most important part of this course (more so that the concrete mathematical results and algorithms covered), and this course is meant to teach student this skill. The fact that the scores for course usefulness is so high indicates that the students themselves understand this. Regarding to the breadth of topics covered, my understanding is that the curriculum has to be kept mostly the same in order to ensure that students can take follow-up courses on algorithms, such as AD and AADS. This is the third year the IDMA course has been given in its new format (changed from the earlier DMFS course), and my assessment is that more and more pieces are falling into place. An indication of this would seem to be that the exam results keep improving (in the sense that more students pass the course). The student evaluations confirm that they perceive the course as well organized, with the different activities creating a logical whole. As last year, my summary is that this is a tough course, but many students enjoy it and learn lots from it.

Hvis relevant, bedes kursusansvarlige kort forholde sig til, hvordan de vigtigste tiltag fra sidste års opfølgning er blevet gennemført, og hvilken effekt de eventuelt har haft.

As in previous years, several adjustments have been made, and most of them seem to have had the intended effect. The student evaluations identify mathematical proofs and formal reasoning as a challenge. This part of the course has been expanded, and the teaching assistants have been encouraged to spend more time on this during the exercise classes. A few more of the detours during the lectures to related problems at the research frontier have been removed, to give some more time to cover material that students will be asked about on the exam. The most important research highlights have been kept, however, and I believe that they serve an important function to show the students

that even material in an introductory course can be connected to cutting-edge research. I believe that such connections should be kept, even if some students complain that the answer to the question "do we need to know this for the exam?" is no. As last year, the teaching assistants were asked this year to spend some of the time during the exercise classes on the board going over solutions to problems, but also to encourage the students to work on problems in advance before they see the solutions. We had no study cafés this year, which was one way of adapting to the decreased number of work hours allocated to the TAs. As in previous years, a lot of effort has been made to put all relevant course information on Absalon, and to send out frequent reminders about the most important points. Most students seem very happy with the current structure of the Absalon course room (except that the material should have been posted slightly earlier and not during the weekend leading up to the course). However, it is also clear from the evaluations that several students still failed to digest this information (claiming that information was only shared close to the exam, when it was in fact available at the beginning of the course).

Refleksion over eksamen (egne og studerendes feedback på eksamen)

It is a bit hard to know how to assess the student feedback on the exam. Quite few students filled in this part of the evaluation, and it stands to reason that the students who feel most strongly about this might be the ones who did not do so well on the exam. There were student complaints that the exam this year was very different from previous years. This would seem to be objectively false. There are always a few new problems (to differentiate which students should get a 10 or 12), but most of the problems on the exam were similar to what students will have seen on the problem sets and on exams from previous years (and for all problem sets and previous exams the students also received detailed worked-out solutions). However, my assessment (shared with the TAs) is that the exam was a bit harder this year than last year, and we have taken this into consideration in the grading. Overall, I am reasonably happy with the exam design. Together with the TAs, I am working on finding better ways of designing problems which

students who knew the material could answer quickly and clearly while students who were less knowledgeable could not just look up the answer in the book (which is a challenge for any introductory course with an open-book exam).

Behov for ændringer i den gældende kursusbeskrivelse? I så fald hvilke?

As in previous years, my personal position is that it seems strange to have open-book exams on introductory courses like this. When I was a BSc student at KTH Royal Institute of Technology back in the 1990s, I think we had roughly the same amount of material but were supposed to learn it by heart. Closed-book exams are still the norm at KTH. However, I get the impression that open-book exams is the established standard at KU. Moving to closed-book exams would probably be such a radical change as to be unimplementable, and so this is nothing that I am suggesting... The biggest concern for next year is that the reduction in the number of teaching assistants, and also teaching assistant hours, hits this course very hard. Since this course is probably the first time most (or all) students on the programs taking IDMA encounter modern, rigorous mathematics (as opposed to the completely different education in calculations that they have had during math lessons in school), it is a bit of a culture shock. I would say that this culture shock is especially pronounced in courses on mathematics, and that teaching assistants are therefore uniquely important for such courses. For IDMA, the teaching assistants play an absolutely crucial role in providing individualized instruction and feedback, and we have been using the allocated TAs and TA hours to the full. But major changes will have to be made for next year, since we now get fewer TAs that are supposed to work fewer hours. We had one TA less for IDMA this year than what we would have had before, which came as a complete surprise. Also, the course makes heavy use of TAs for exercise classes and compulsory hand-in problems with grading and resubmission. We see that this is improving results (and that the student evaluations point this out as a key reason for the success of the course), but since the number of hours the TAs are assigned has decreased significantly, we will need to make major changes in the course set-up (to the detriment of the students and the course results). This was already a serious problem this

year, but the situation was saved by that the TAs got 120 hours. For next year, based on current information it seems we are looking at a scenario with (1) significant increase in number of students, (2) decrease in number of TAs, and (3) decrease in the number of hours these TAs are paid, and as things stand now this will create problems that are very hard to manage without causing TA and instructor overload. The one solution I can see is to cut down on TA manning of exercise classes, but I am seriously concerned that we will see a corresponding drop in student performance.

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